Tutorial –II
Kraft-McMillan Inequality

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Problem No: 1

Consider the four codes listed below:

Symbol	Code I	Code II	Code III	Code IV
s_0	0	0	0	00
s_1	10	01	01	01
s_2	110	001	011	10
s_3	1110	0010	110	110
s_4	1111	0011	111	111

Problem No: 1 Cont..

3) Kraft–McMillan check (binary): compute ∑ 2^(−ℓ_i)

Compute lengths ℓ and the sum step-by-step (digit-by-digit):

- **Code I** lengths `1,2,3,4,4`
  - `2⁻¹ = 1/2 = 0.5`
  - `+ 2⁻² = 1/4 = 0.25 → 0.75`
  - `+ 2⁻³ = 1/8 = 0.125 → 0.875`
  - `+ 2⁻⁴ = 1/16 = 0.0625 → 0.9375`
  - `+ 2⁻⁴ = 0.0625 → 1.0000`
  - **Sum = 1.0 ≤ 1** (equality).
- **Code II** lengths `1,2,3,4,4` (same as I) → **Sum = 1.0 ≤ 1.**
- **Code III** lengths `1,2,3,3,3`
  - `1/2 = 0.5`
  - `+1/4 = 0.25 → 0.75`
  - `+1/8 = 0.125 → 0.875`
  - `+1/8 = 0.125 → 1.000`
  - `+1/8 = 0.125 → 1.125`
  - **Sum = 1.125 > 1** (violates Kraft).
- **Code IV** lengths `2,2,2,3,3`
  - `3×2⁻² = 3×1/4 = 0.75`
  - `+ 2×2⁻³ = 2×1/8 = 0.25 → 1.00`
  - **Sum = 1.0 ≤ 1** (equality).

Answer the following:

- Two of these four codes are prefix codes. Identify them, and construct their individual decision trees.
- Apply the Kraft-McMillan In equality to codes I, II, III, and IV. Discuss your results in light of those obtained in part (a).